MDPI

Article

Multi-Granularity Feature Decomposition with Ranking-Aware Regularization for Old Film Restoration

Feifan Cai [1,2], Qi Zhang [1,2], Kun Yang [3], Lei Chen [3] and Youdong Ding [1,2,*]

[1] Shanghai Film Academy, Shanghai University, 788 Guangzhong Road, Shanghai 200072, China; feifan@shu.edu.cn (F.C.); zq10501@shu.edu.cn (Q.Z.)
[2] Shanghai Engineering Research Center of Motion Picture Special Effects, Shanghai 200072, China
[3] College of Mechanical Engineering, Taiyuan University of Technology, Taiyuan 030024, China; yangkun@tyut.edu.cn (K.Y.); chenlei839@163.com (L.C.)
* Correspondence: ydding@shu.edu.cn (Y.D.)

Abstract

Old film restoration remains difficult because archival footage often contains scratches, blotches, flicker, blur, and temporal artifacts that act at different spatial and temporal scales. These mixed degradations make structural recovery, regional texture repair, and fine artifact suppression compete within a restoration model. We propose MgfrOFR, a multi-granularity extension of a Mamba-based old film restoration backbone. MgfrOFR decomposes propagated features into coarse-to-fine restoration branches and feeds branch summaries back through channel-wise recalibration, so that different degradation scales can be represented without relying on a single feature stream. It also introduces a ranking-aware diversity regularizer (RATR) that penalizes correlated temporal-similarity rankings across branches during training while preserving the baseline inference interface. Experiments on the SRWOV benchmark show statistically significant full-reference gains on synthetic clips, the best NIQE score and an improved BRISQUE score relative to MambaOFR on real-world archival footage, while CLIP-IQA+ remains below the enhancement-oriented OldPhoto baseline and the MambaOFR backbone. These results support the narrower claim that balancing coarse-to-fine feature branches with ranking-aware regularization improves old film restoration under heterogeneous degradation.

Keywords: old film restoration; multi-granularity feature decomposition; ranking-aware regularization; state space models; feature-space symmetry

1. Introduction

Old film restoration must recover visual content from footage affected by multiple interacting degradation types. Archival film stock often contains scratches, dust, blotches, flicker, blur, grain, and fading within the same clip, and these defects continue to accumulate as the original material deteriorates over time. Digital restoration therefore plays an important role in preserving cinematic heritage, historical documentaries, and large archival collections [1]. The technical difficulty is that these degradations occupy different spatial and temporal scales: large missing regions require global context, mid-scale stains require regional repair, and thin scratches or dust require local precision. In this paper, symmetry is used in a precise representation-level sense, not as visible geometric symmetry. The restoration evidence is organized into complementary granularity branches, and the learning objective is required to stay invariant under any permutation of these branches. This branch-permutation invariance acts as a symmetric constraint: it prevents optimization

Received:
Revised:
Accepted:
Published:

Copyright: © 2026 by the authors. Submitted to *Symmetry* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license.

from favoring one dominant cue and keeps coarse structural recovery, regional texture repair, and fine artifact suppression balanced under spatially and temporally asymmetric damage. Permutation invariance is a standard symmetry principle in machine learning, which links this design to symmetry-oriented artificial intelligence.

Learning-based old film restoration (OFR) has improved substantially, but existing propagation pipelines still tend to mix different restoration evidence in one latent stream. RTN introduced bidirectional recurrent propagation with learnable defect masking [2], and RRTN strengthened alignment with recursive transformer propagation [3]. MambaOFR adopted state space models for OFR, improved efficiency, and established the SRWOV benchmark [4–6]. These methods provide strong temporal modeling, yet structural recovery and fine artifact suppression remain jointly encoded in propagated features. When heterogeneous evidence competes in one latent representation, coarse repair can overwrite fine evidence, while aggressive local artifact suppression can distort global structure.

The gap addressed in this work is the absence of an OFR-specific mechanism that separates restoration evidence by granularity and keeps the resulting branches complementary. Multi-granularity designs have been effective in detection and recognition [7–11], but those settings use different supervision and feature semantics. A direct multi-branch extension for OFR may also collapse into correlated branches unless diversity is explicitly regularized.

To address this limitation, we propose MgfrOFR, which augments MambaOFR with a Multi-Granularity Feature Extractor (MGFE) and a ranking-aware diversity regularizer (RATR). MGFE reorganizes propagated Mamba features into coarse-to-fine restoration branches, allowing structural completion, regional repair, and fine-detail suppression to be represented by dedicated descriptors rather than a single hidden stream. RATR further penalizes correlated temporal-similarity rankings across branches during training, reducing redundant branch behavior without changing the baseline inference interface. Under a unified reproduction protocol on SRWOV, MgfrOFR retains the efficient MambaOFR backbone while adding explicit granularity control with modest overhead. Recent Mamba-based restoration methods mainly improve the scanning design inside the backbone; MgfrOFR instead leaves the propagation backbone unchanged and operates on its output features, so the granularity decomposition and ranking diversity stay orthogonal to the choice of state-space backbone. This keeps the contribution architecturally non-invasive: MGFE is a shape-preserving insertion at inference, and ranking diversity is confined to training. The main contributions are summarized as follows:

1. We formulate old film restoration as a representation-level branch-permutation symmetry problem. Coarse structural recovery, regional texture repair, and fine artifact suppression are treated as complementary feature components whose diversity objective stays invariant under branch permutation, so the branches remain balanced under degradation-induced spatial and temporal asymmetry.
2. We design a multi-granularity feature decomposition mechanism for Mamba-based old film restoration. MGFE inserts learnable restoration tokens into propagated features, progressively aggregates them into coarse-to-fine branches, and feeds branch summaries back to the spatial stream through channel-wise recalibration.
3. We introduce a ranking-aware diversity regularizer (RATR) to improve branch complementarity during training. By penalizing correlated temporal-similarity rankings across granularity branches with differentiable rank correlation, RATR reduces redundant branch behavior while preserving the original baseline inference interface.
4. We conduct a unified evaluation on the SRWOV benchmark covering synthetic and real-world archival footage. The results, statistical tests, ablation studies, and complexity analysis show that MgfrOFR improves restoration fidelity and no-reference

naturalness with modest additional computational cost.

2. Related Work

2.1. Old Film and Video Restoration

Old film restoration differs from general video restoration because it must jointly address structured artifacts, photometric degradation, and quality degeneration. Scratches, dust, and blotches require defect-aware repair, while flicker, grain, fading, blur, and noise require photometric and temporal correction. Traditional approaches decompose the problem into detection and inpainting stages, using hand-crafted features with interpolation-based filling [12] or sparse collaborative filtering [13,14]. Deep learning shifted OFR toward end-to-end restoration: DeepRemaster [1] processes spatiotemporal volumes with 3D convolutions, RTN [2] introduces bidirectional recurrent propagation with a learnable soft defect mask, RRTN [3] strengthens alignment with recursive transformer networks and second-order grid propagation, and MambaOFR [4] uses MambaIR-style [6] state-space modules while introducing the SRWOV benchmark.

Existing video restoration and enhancement systems provide useful context but are not tailored to OFR-specific defect structure. General video restoration methods such as EDVR [15], BasicVSR [16], BasicVSR++ [17], VRT [18], and RVRT [19] model temporal information effectively, but they do not explicitly handle old-film defects. Diffusion and foundation-prior systems, including DiffBIR [20], SUPIR [21], and SeeSR [22], improve perceptual detail for blind image restoration or real-world super-resolution, but they are not designed around temporal old-film artifacts or SRWOV-style video protocols. This leaves a specific need for propagation features that can separate structural completion, regional repair, and fine artifact suppression.

2.2. State Space Models for Vision

State space models offer an efficient alternative to attention-heavy restoration backbones. The computational cost of self-attention scales quadratically with sequence length, which is costly for high-resolution video. Structured state space sequence models [23] reformulate sequence modeling through continuous-time state transitions, reducing complexity from $O(L^2)$ to $O(L)$ while maintaining a global receptive field. Mamba [5] further introduces selective state transitions with hardware-aware recurrence. In vision, MambaIR [6] demonstrates 2D selective scanning for image restoration, VMamba [24] builds hierarchical feature pyramids with four scanning routes, and VSRM [25] alternates spatial and temporal Mamba blocks for video super-resolution at linear cost.

These properties are well suited to OFR, which combines long temporal dependency, large spatial damage, and high-resolution input. MambaOFR [4] shows that Mamba-based propagation can improve both efficiency and restoration quality relative to transformer counterparts. However, long-range aggregation alone does not specify which degradation scale each feature should represent. Recent Mamba-based restoration methods improve this backbone mainly by redesigning the scan path, such as the multi-route scanning of VMamba [24] and the alternating spatial-temporal scan blocks of VSRM [25]. MgfrOFR is orthogonal to this line of work: it keeps the MambaOFR scan and propagation unchanged and instead organizes the already propagated features into granularity branches with a training-time ranking constraint. The increment over MambaOFR is therefore an explicit granularity-decomposition and branch-diversity mechanism rather than a new scanning operator, and it transfers to any state-space backbone that exposes propagated features.

2.3. Multi-Scale and Multi-Branch Representations

Multi-scale and multi-branch representations are useful when one feature scale cannot capture both global context and fine local evidence. Feature Pyramid Networks [7] and HRNet [8] show the value of explicit multi-scale decomposition for dense prediction, while MIRNet-v2 [26] shows that complementary multi-scale context can improve low-level reconstruction. In person re-identification, PCB [9] decomposes features along spatial stripes, MGN [10] combines global and local branches at different partition granularities, and ReIDMamba [11] adapts this idea to state space models. These studies motivate branch-structured representations, but they differ from MGFE in both mechanism and objective. Multi-scale pyramids build features at several spatial resolutions and fuse them across scales, whereas MGFE keeps a single full-resolution propagated feature map and separates granularity through the fusion rate of inserted restoration tokens, adding no extra resolution branch to the reconstruction path. Token-based and stripe-based recognition models read their partitioned features directly into the task head, whereas MGFE returns its branch summaries to the spatial stream through channel-wise recalibration and exposes them only as ranking targets during training. Their supervision also targets recognition rather than video restoration.

2.4. Diversity and Ranking Regularization

Branch structure is useful only when the branches capture complementary cues rather than correlated copies of the same evidence. Contrastive learning [27] provides a general diversity mechanism, but it does not directly regulate pairwise ordering behavior across branches. Two branches may differ in magnitude while ranking all candidate frames in the same order, producing apparent feature diversity but redundant temporal structure. Differentiable Kendall's Tau [28] enables gradient-based penalization of correlated rankings, which is directly applicable to flow-guided temporal correspondences in OFR.

2.5. Representation Symmetry in Restoration

For this paper, representation symmetry denotes balanced and comparable organization of complementary restoration evidence rather than visible geometric symmetry. A learned representation becomes unbalanced when optimization favors one dominant cue and suppresses other cues needed for restoration. Feature-fusion and correlation-based representation studies show that multi-view or multi-branch features require explicit organization rather than simple concatenation [29,30], while adjacent image-processing studies use attention, detection, or differential-equation priors to organize heterogeneous visual evidence for anomaly detection, object detection, segmentation, and denoising [31–33]. Feature-wise modulation provides a practical way to inject conditioning signals into image-processing networks [34]. This restricted view is relevant to OFR because film damage is uneven across both spatial regions and frames.

MgfrOFR adapts these ideas to restoration rather than recognition. First, branch tokens are inserted into propagated restoration features rather than identity-oriented embeddings. Second, branch summaries are fed back into the restoration stream through channel-wise recalibration, so they affect reconstruction rather than only auxiliary supervision. Third, positives and negatives are defined by temporal correspondence within and across clips rather than person identities. The resulting design addresses two OFR-specific gaps: the lack of explicit multi-granularity decomposition within the temporal propagation pipeline, and the lack of diversity regularization that keeps parallel restoration branches complementary, as detailed in Section 3.

2.6. Positioning of MgfrOFR

Because the contribution of MgfrOFR is an increment over the MambaOFR backbone rather than a new restoration architecture, we make the distinction explicit along architectural and theoretical axes. Table 1 contrasts MgfrOFR with the four most related families: multi-scale pyramids, token- or stripe-based recognition models, recent Mamba-based restoration backbones, and diversity or ranking regularization. Multi-scale pyramids and Mamba-restoration methods change *how features are produced*—by adding resolution branches or by redesigning the scan path—whereas MgfrOFR leaves the propagation backbone and its scan unchanged and instead reorganizes the already propagated features by token-fusion granularity at a single resolution. Token- and stripe-based recognition models read their partitioned features directly into a task head, whereas MgfrOFR returns its branch summaries to the spatial stream through channel-wise recalibration and exposes them only as ranking targets during training. Finally, unlike generic diversity losses that act on feature magnitude, RATR regularizes the *ranking* behavior of branches and is accompanied by the permutation- and direction-exchange invariances of Section 3.7. The increment over MambaOFR is therefore an explicit granularity-decomposition and branch-diversity mechanism that is orthogonal to the choice of state-space backbone, rather than a new scanning operator.

Table 1. Positioning of MgfrOFR against the most related multi-scale, token/stripe recognition, Mamba-restoration, and diversity-regularization families along architectural and theoretical axes.

Aspect	Multi-scale pyramids	Token/stripe ReID	Mamba restoration	MgfrOFR (Ours)
Granularity source	Multiple spatial resolutions	Spatial partitions/stripes	Single scan stream	Token-fusion rate at one resolution
Branch diversity	Implicit via fusion	None explicit	None	Ranking-aware regularization
Invariance/symmetry	–	–	–	Branch permutation + direction exchange
Feedback to features	Cross-scale fusion	Read by task head	–	Channel-wise recalibration
Inference cost	Extra resolution branches	Partition heads	Backbone only	Shape-preserving; RATR training-only
Target task	Dense prediction	Person re-identification	Image/video restoration	Old film restoration

3. Methods

3.1. Problem Setup

Given a degraded old-film clip $X = \{x_t\}_{t=1}^{T}$, the task is to estimate a restored clip $Y = \{y_t\}_{t=1}^{T}$ that recovers image content while suppressing structured film defects and photometric degradation. Synthetic SRWOV samples provide paired clean frames for direct supervision. Real-world archival clips do not provide ground truth, so their evaluation relies on no-reference quality metrics and qualitative inspection. The method must therefore preserve temporal consistency while handling degradations that vary across both spatial

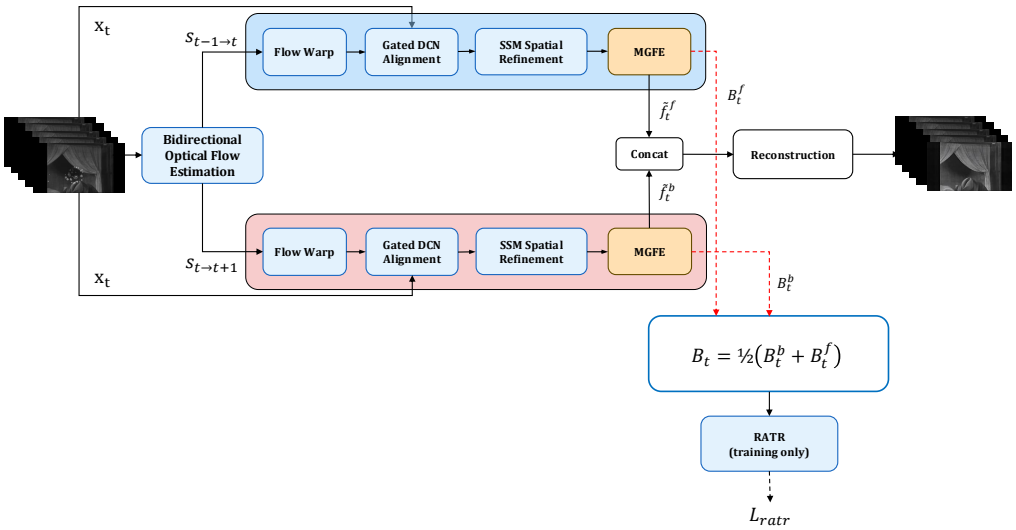


Figure 1. MgfrOFR framework. MGFE decomposes propagated features into G=3 multi-granularity restoration-token branches and uses their summaries for channel-wise recalibration. RATR is used only during training (dashed red path) to penalize redundant temporal-ranking correlations and produce $\mathcal{L}_{\text{intra}}$ and $\mathcal{L}_{\text{inter}}$.

scale and time.

3.2. Input Representation

MgfrOFR operates on the propagated feature maps produced by the baseline bidirectional restoration backbone. For each time step t, the backward and forward streams produce aligned feature maps $f_t^b, f_t^f \in \mathbb{R}^{C_f \times H \times W}$ after motion-aware propagation. MgfrOFR treats these maps as the only interface to the backbone: MGFE reads each propagated map, returns a recalibrated map with the same spatial size and channel count, and emits branch descriptors used by RATR during training.

MgfrOFR follows the same bidirectional propagation framework as MambaOFR [4] while adding an explicit coarse-to-fine representation pipeline. Given a degraded input clip $X = \{x_t\}_{t=1}^{T}$, the baseline first propagates content in backward and forward directions so that neighboring observations are motion-aligned before reconstruction. MgfrOFR then separates restoration evidence by spatial granularity and regularizes the resulting branches during training. In the representation-level symmetry framing used here, coarse, medium, and fine restoration evidence should remain balanced even when the input damage is spatially and temporally asymmetric.

The two added components serve different roles. MGFE converts each propagated feature map into a token sequence, inserts learnable restoration tokens, derives branch descriptors from progressively fused token groups, and feeds their summaries back through channel-wise recalibration. RATR is used only during training; it discourages redundant ranking behavior across branches and does not add an inference module. The reconstruction head follows the original MambaOFR contract and outputs the restored frame. The overall architecture is illustrated in Figure 1.

Algorithm 1 MgfrOFR forward pass for a clip of length T.

Require: Degraded clip $X = \{x_t\}_{t=1}^{T}$
Ensure: Restored frames $\{y_t\}_{t=1}^{T}$
Initialize descriptor buffer $\mathcal{B} \leftarrow \emptyset$
for $t = 1$ to T **do**
  $f_t^b, f_t^f \leftarrow \text{Backbone}(X, t)$ ▷ Bidirectional propagation
  for $d \in \{b, f\}$ **do**
    $\widetilde{f}_t^d, B_t^d \leftarrow \text{MGFE}(f_t^d)$ ▷ Multi-granularity decomposition
  end for
  $B_t \leftarrow \frac{1}{2}(B_t^b + B_t^f)$ ▷ Average branch descriptors
  $\mathcal{B} \leftarrow \mathcal{B} \cup \{B_t\}$ ▷ Accumulate into batch buffer
  $y_t \leftarrow \mathcal{R}([\widetilde{f}_t^b; \widetilde{f}_t^f], x_t)$ ▷ Reconstruction
end for
if training **then**
  $\mathcal{L}_{\text{ratr}} \leftarrow \text{RATR}(\{B_t\}_{t=1}^{T})$ ▷ Applied on full batch after accumulation
  $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{pix}} + \lambda_{\text{perc}}\mathcal{L}_{\text{perc}} + \lambda_{\text{gan}}\mathcal{L}_{\text{gan}} + \lambda_{\text{ratr}}\mathcal{L}_{\text{ratr}}$
end if

3.3. Pipeline Overview

For each time step t, the backward and forward propagation paths of the baseline produce feature maps $f_t^b, f_t^f \in \mathbb{R}^{C_f \times H \times W}$. MGFE transforms each propagated feature into a recalibrated feature map and a set of branch descriptors:

$$\left(\widetilde{f}_t^d, B_t^d\right) = \text{MGFE}\left(f_t^d\right), \quad d \in \{b, f\}, \tag{1}$$

where d indexes the propagation direction, $B_t^d \in \mathbb{R}^{G \times D}$ is the branch-descriptor tensor and $D = C_f$ in this work. The reconstruction head keeps the original MambaOFR interface and combines the two recalibrated directional features:

$$y_t = \mathcal{R}\left([\widetilde{f}_t^b; \widetilde{f}_t^f], x_t\right), \qquad B_t = \frac{1}{2}\left(B_t^b + B_t^f\right), \tag{2}$$

where $\mathcal{R}(\cdot)$ denotes the unchanged reconstruction head of MambaOFR and $[\cdot;\cdot]$ denotes channel concatenation. Equation (2) matches the implemented data flow: the directional descriptors are averaged before RATR is applied, so the diversity constraint acts on one descriptor tensor per frame rather than duplicating a similar constraint on two directional streams. Figure 1 illustrates the same data flow, and Algorithm 1 summarizes the forward pass.

3.4. Baseline Interface and Preserved Components

MgfrOFR preserves the MambaOFR backbone, bidirectional propagation order, reconstruction head, and training loop. The only inference-time insertion is MGFE, which maps f_t^d to $\widetilde{f}_t^d$ without changing tensor shape. RATR is an auxiliary training loss over branch descriptors and is disabled during inference. This interface keeps the baseline reconstruction contract unchanged: the head still receives concatenated backward and forward propagated features together with the current degraded frame.

3.5. Multi-Granularity Feature Extractor (MGFE)

MGFE separates restoration evidence by spatial granularity while keeping the propagated feature map as the spatial carrier for reconstruction. Large missing regions, mid-scale blotches, and thin scratches require different aggregation ranges, but a single hidden tensor does not require these cues to remain separable. MGFE addresses this by exposing multiple restoration-token branches with different aggregation rates and feeding their summaries

back as a recalibration signal. This component provides the branch-balance mechanism of MgfrOFR: each degradation scale receives a dedicated descriptor while reconstruction still uses a common spatial feature carrier. Figure 2 shows the detailed design.

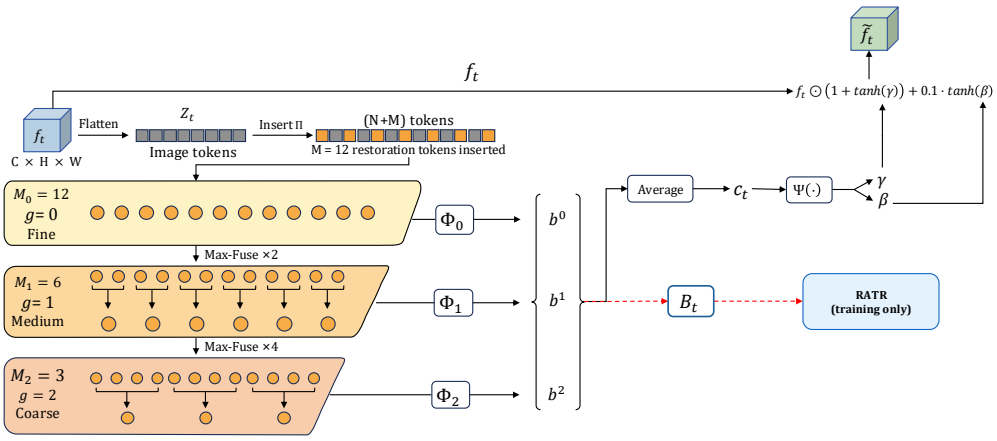


Figure 2. MGFE module. M=12 learnable restoration tokens are inserted into image tokens, fused by progressive average fusion, and summarized into G=3 restoration-token branches. The resulting descriptors generate scale and shift parameters for channel-wise recalibration.

Since MGFE processes each propagation direction identically, we drop the direction superscript d and write $f_t \in \mathbb{R}^{C_f \times H \times W}$ for brevity. We first flatten it into an image-token sequence

$$Z_t = \mathrm{Flat}(f_t) \in \mathbb{R}^{N \times C_f}, \qquad N = HW, \tag{3}$$

where $\mathrm{Flat}(\cdot)$ preserves raster order and stacks spatial positions into token form. We then append M learnable restoration tokens $R \in \mathbb{R}^{M \times C_f}$, initialized from a truncated normal distribution with standard deviation 0.02, and insert them into the image-token stream through an operator $\Pi(\cdot,\cdot)$:

$$\widehat{Z}_t = \Pi(Z_t, R) \in \mathbb{R}^{(N+M) \times C_f}. \tag{4}$$

The insertion operator Π distributes restoration tokens uniformly across the image-token stream. It partitions the N image tokens into $M{+}1$ approximately equal chunks and places one restoration token after each of the first M chunks while preserving raster order. Each chunk receives $\lfloor N/(M{+}1) \rfloor$ or $\lceil N/(M{+}1) \rceil$ image tokens so that the total remains exactly N. For example, with $M = 3$ and $N = 9$, the four chunks contain $\{3,2,2,2\}$ image tokens, and the three restoration tokens are inserted at cumulative positions $\{3,6,9\}$ within the 12-token interleaved sequence. This deterministic interleaving gives restoration tokens balanced coverage over the full spatial support.

MGFE builds G granularity branches by progressively fusing restoration tokens. The fusion scheme follows ReIDMamba [11] and is adapted here to propagated spatial restoration features. To make every branch well defined, the number of restoration tokens must satisfy the divisibility constraint

$$M = k \cdot 2^{G-1}, \quad k \in \mathbb{N}^+, \tag{5}$$

so that every branch size $M_g = M/2^g$ is an integer for $g \in \{0, \ldots, G-1\}$. We use $G = 3$

and $M = 12$ ($k = 3$) throughout; see Section 4.5 for alternative configurations.

For branch $g \in \{0, \ldots, G-1\}$, MGFE fuses restoration tokens with a sampling rate of 2^g. The fused restoration tokens are defined as

$$R^{(g)}[j] = \mathcal{F}\Big(\{R[j{\cdot}2^g + k]\}_{k=0}^{2^g-1}\Big), \quad j = 0, \ldots, M_g - 1, \tag{6}$$

where $\mathcal{F}(\cdot)$ denotes an element-wise average operator over all tokens in the fusion group:

$$\mathcal{F}\Big(\{R[j \cdot 2^g + k]\}_{k=0}^{2^g-1}\Big) = \frac{1}{2^g} \sum_{k=0}^{2^g-1} R[j \cdot 2^g + k] \ \in \mathbb{R}^{C_f}. \tag{7}$$

The average is taken channel-wise. We use average fusion as the default because it preserves a stable aggregate of degradation evidence across tokens and avoids over-reliance on a single extreme activation. Max fusion is evaluated as an alternative in the ablation study (Section 4.4); while it can preserve sparse high-contrast activations, the sensitivity analysis shows that average fusion gives a better full-reference and perceptual balance for the tested restoration setting. Smaller g values preserve more tokens and retain finer local detail, while larger g values compress wider support and emphasize coarse restoration context.

The fused tokens are re-inserted into the same image-token sequence using the same uniform-spacing insertion rule:

$$\widehat{Z}_t^{(g)} = \Pi\Big(Z_t, R^{(g)}\Big), \tag{8}$$

and the resulting sequence is processed by a lightweight branch-specific token mixer $\Phi_g(\cdot)$:

$$U_t^{(g)} = \Phi_g\Big(\widehat{Z}_t^{(g)}\Big) \in \mathbb{R}^{(N+M_g)\times C_f}, \tag{9}$$

where $M_g = M/2^g$. Each Φ_g consists of two sequential linear-complexity token blocks. Each block contains a context-augmented projection sublayer and a feed-forward sublayer. The projection sublayer applies LayerNorm, computes a global context vector by mean pooling over the token dimension, adds the projected context to the per-token linear projection, and applies a residual connection. The feed-forward sublayer applies LayerNorm followed by a two-layer MLP with GELU activation and hidden dimension $2C_f$, again with a residual connection. Formally, for input tokens $\mathbf{h}$, each block computes

$$\mathbf{h}' = \mathbf{h} + W_t\,\mathrm{LN}(\mathbf{h}) + W_c\,\overline{\mathrm{LN}(\mathbf{h})}, \qquad \mathbf{h}'' = \mathbf{h}' + \mathrm{MLP}(\mathrm{LN}(\mathbf{h}')), \tag{10}$$

where $\overline{(\cdot)}$ denotes mean pooling over the token dimension, $W_t, W_c \in \mathbb{R}^{C_f \times C_f}$ are learned projections, and MLP consists of $\mathrm{Linear}(C_f \to 2C_f) \to \mathrm{GELU} \to \mathrm{Linear}(2C_f \to C_f)$. The global context term $W_c\,\overline{\mathrm{LN}(\mathbf{h})}$ gives each token a lightweight full-sequence summary at linear cost, avoiding the quadratic complexity of self-attention while retaining inter-token communication. Each branch has its own pair of token blocks, allowing branch-specific parameterization with modest overhead.

After branch processing, MGFE summarizes each branch by averaging the updated restoration-token positions:

$$b_t^{(g)} = \frac{1}{M_g} \sum_{m=1}^{M_g} U_t^{(g)}[\pi_m^{(M_g)}] \in \mathbb{R}^D, \tag{11}$$

where $\{\pi_m^{(M_g)}\}_{m=1}^{M_g}$ denote the uniformly spaced insertion positions for M_g tokens,

and $D = C_f$. Stacking all branch descriptors gives

$$B_t = \left[b_t^{(0)}, b_t^{(1)}, \ldots, b_t^{(G-1)} \right]^\top \in \mathbb{R}^{G \times D}. \tag{12}$$

Because the descriptors are extracted from a common propagated feature under different token-fusion rates, they share semantic alignment while differing in granularity. This descriptor tensor B_t is the input to RATR.

MGFE also converts the branch descriptors into a channel-wise recalibration signal. We first average the branch context,

$$c_t = \frac{1}{G} \sum_{g=0}^{G-1} b_t^{(g)}, \qquad [\gamma_t, \beta_t] = \Psi(c_t), \tag{13}$$

where $\Psi(\cdot)$ is a projection head consisting of LayerNorm $\to$ Linear$(C_f \to 2C_f)$ $\to$ GELU $\to$ Linear$(2C_f \to 2C_f)$ that outputs concatenated scale and shift vectors. The recalibrated propagated feature is then

$$\widetilde{f}_t = f_t \odot (1 + \tanh(\gamma_t)) + 0.1 \, \tanh(\beta_t), \tag{14}$$

where the scale and shift vectors from Equation (13) are broadcast over the spatial dimensions. Both terms have bounded, interpretable ranges because $\tanh(\cdot) \in (-1, 1)$. The multiplicative gate satisfies $1 + \tanh(\gamma_t) \in (0, 2)$ and equals 1 when $\gamma_t \approx \mathbf{0}$ at initialization, so recalibration begins as an identity scaling. The additive term is hard-bounded elementwise, $|0.1 \, \tanh(\beta_t)| < 0.1$, independently of the magnitude of β_t. This bound is the precise meaning of the "10% modulation": the additive shift of each channel never exceeds 0.1 in absolute value, which is 10% of the unit reference gain of the multiplicative branch at initialization. Standard FiLM modulation [34] is unbounded; this explicit bound instead keeps the recalibrated feature close to the original propagated feature when the branch context is uncertain. Once the branch descriptors become informative, coarse branches can supply global structural preferences and fine branches can emphasize localized damage patterns.

In summary, MGFE separates restoration evidence by spatial granularity through lightweight token mixing while preserving the baseline interface through residual-style recalibration. The branch descriptors B_t and the recalibration signal (γ_t, β_t) serve two different roles: the descriptors provide ranking targets for RATR (Section 3.6), while the recalibration feeds granularity-aware modulation back into the propagated feature. This separation allows ablation studies to disable recalibration while keeping branch descriptors active for diversity regularization (see the RATR-only configuration in Section 4.4). Explicit branches alone, however, do not guarantee complementary features. If all branches induce the same sample rankings, branch balance degenerates into redundant descriptors. This motivates the training regularizer described next.

3.6. Ranking-Aware Diversity Regularization (RATR)

RATR discourages branch collapse by penalizing excessive ranking correlation between branches. A simple diversity loss could force descriptors to be orthogonal, but orthogonality does not ensure that branches organize restoration evidence differently across samples. The relevant behavior is the branch-specific similarity order induced when an anchor frame is compared with nearby positives and harder negatives. If different branches rank the same candidates identically, they remain redundant even if their descriptor values differ. RATR therefore formulates diversity as an anchor-centered ranking problem using Kendall's [35] rank correlation coefficient, made differentiable by DiffKendall [28]. Figure 3 illustrates the workflow.

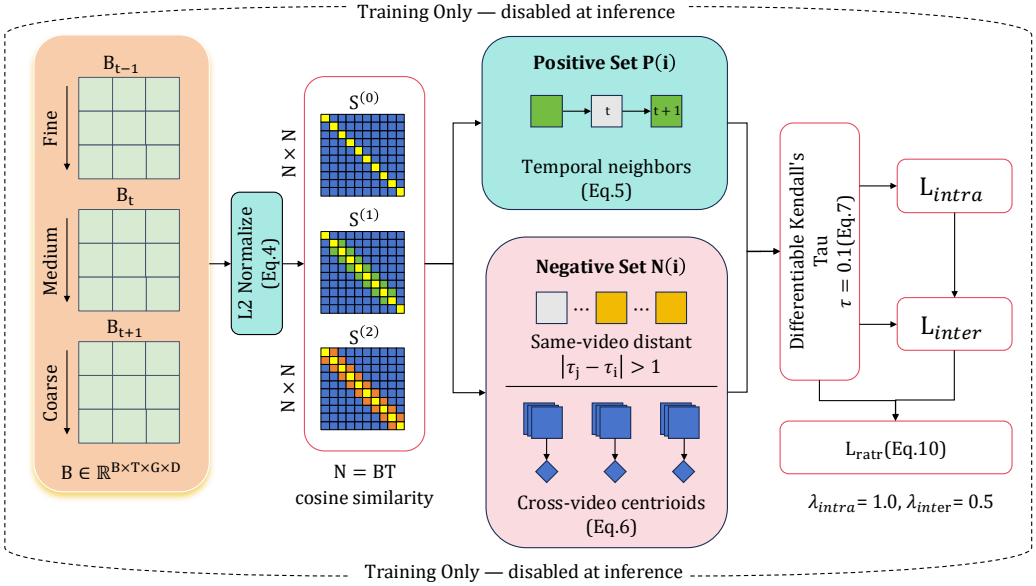


Figure 3. RATR module. Differentiable Kendall's Tau compares branch-specific ranking sequences from temporal correspondences. Neighbor frames define positives, while distant same-video frames and cross-video centroids define negatives, producing $\mathcal{L}_{\text{intra}}$ and $\mathcal{L}_{\text{inter}}$ during training only.

Let the averaged branch descriptors from Equation (2) be organized as a 4D tensor of shape $N_b \times T \times G \times D$, where N_b is the batch size. We flatten batch and time into $\bar{N} = N_b T$ samples and normalize the descriptors branch-wise:

$$\hat{b}_i^{(g)} = \frac{b_i^{(g)}}{\|b_i^{(g)}\|_2}, \qquad i = 1, \ldots, \bar{N}. \tag{15}$$

For each branch g, we compute a cosine-similarity matrix

$$\mathcal{S}^{(g)}[i,j] = \left\langle \hat{b}_i^{(g)}, \hat{b}_j^{(g)} \right\rangle. \tag{16}$$

This matrix is not used as a standalone supervision target. Instead, it provides the positive and negative ranking sequences that form the anchor-positive-negative structure regularized by RATR.

For anchor i, the positive set is defined from temporal correspondences within the same clip:

$$\mathcal{P}(i) = \{ j \mid v_j = v_i,\ |t_j - t_i| = 1 \}, \tag{17}$$

where v_i and t_i denote the clip index and frame index of sample i. This temporal adjacency criterion is justified because the flow-guided deformable alignment in MambaOFR has already warped neighboring frames toward the anchor frame before MGFE is applied. After this alignment, adjacent frames are the most reliable proxy for true flow-based correspondences in the feature domain. Since $|\mathcal{P}(i)| \leq 2$, boundary frames provide only one positive neighbor. We therefore evaluate the intra-ranking term only on anchors with at least two positives:

$$\mathcal{I}_{\text{intra}} = \{ i \mid |\mathcal{P}(i)| \geq 2 \}. \tag{18}$$

For the seven-frame clips used in training, this excludes only the first and last frame of each clip. Interior anchors then contribute exactly one pairwise ordering comparison to $\mathcal{L}_{\text{intra}}$. Restricting the positive set to the two immediate neighbors is a deliberate accuracy-over-quantity choice: after flow-guided alignment, $t \pm 1$ are the correspondences with the highest sub-pixel reliability, whereas more distant frames accumulate alignment error and would

inject label noise into the ranking targets. The signal is nonetheless sufficient for stable regularization for three reasons. First, two neighbors already define a complete pairwise order around the anchor, which is the minimal unbiased unit that a rank-correlation penalty requires. Second, $\mathcal{L}_{\text{intra}}$ is averaged over all interior anchors in the batch, so each per-anchor estimate is weak but the batch-level gradient is stable. Third, $\mathcal{L}_{\text{inter}}$ supplies a complementary, higher-cardinality ranking signal from the richer negative set, covering the long-range ordering that the local positives omit.

Negative evidence is drawn from two harder sources: temporally distant frames inside the same clip and clip-level representatives from other clips. The same-video negatives are

$$\mathcal{N}_{sv}(i) = \{ j \mid v_j = v_i,\ |t_j - t_i| > 1 \}, \tag{19}$$

and the clip-level centroid for another video $v \neq v_i$ is

$$\bar{b}_v^{(g)} = \frac{1}{|\mathcal{T}_v|} \sum_{t \in \mathcal{T}_v} \widehat{b}_{(v,t)}^{(g)}, \tag{20}$$

where $\mathcal{T}_v$ denotes the set of frame indices in clip v. We then construct the negative ranking sequence as

$$\bar{\mathcal{S}}_i^{(g)} = \Big[\mathcal{S}^{(g)}[i,j] \Big]_{j \in \mathcal{N}_{sv}(i)} \oplus \Big[\big\langle \widehat{b}_i^{(g)}, \bar{b}_v^{(g)} \big\rangle \Big]_{v \neq v_i}, \tag{21}$$

where $\oplus$ denotes sequence concatenation. Same-video distant frames are hard negatives because they share content and degradation statistics with the anchor but are no longer immediate temporal neighbors. Off-video centroids provide a low-variance clip-level representative that avoids the $O(\bar{N}^2)$ cost of enumerating every cross-clip frame pair while still capturing inter-clip distributional contrast.

To compare how two branches rank the same candidate set, we adopt a differentiable version of Kendall's Tau. For sequences $p, q \in \mathbb{R}^L$, the differentiable statistic is

$$\mathrm{DKTau}(p, q; \tau) = \frac{1}{\binom{L}{2}} \sum_{1 \le u < v \le L} \tanh\!\left(\frac{p_u - p_v}{\tau} \right) \cdot \tanh\!\left(\frac{q_u - q_v}{\tau} \right), \tag{22}$$

where $\tau > 0$ is a temperature parameter. As $\tau \to 0$, Equation (22) approaches the hard sign comparison used in the exact Kendall coefficient; as τ becomes too large, pairwise order differences are oversmoothed and gradients vanish. We set $\tau = 0.1$ (see Section 4.5 for the sensitivity to this choice). A lower value of $\mathrm{DKTau}(p, q; \tau)$ indicates weaker ranking correlation between branches and therefore stronger branch complementarity.

Using Equation (22), we compute the positive-ranking correlation loss as

$$\mathcal{L}_{\text{intra}} = \frac{1}{|\mathcal{I}_{\text{intra}}| \binom{G}{2}} \sum_{i \in \mathcal{I}_{\text{intra}}} \sum_{g < g'} \mathrm{DKTau}\Big(\mathcal{S}^{(g)}[i, \mathcal{P}(i)],\ \mathcal{S}^{(g')}[i, \mathcal{P}(i)];\ \tau \Big), \tag{23}$$

and the negative-ranking correlation loss as

$$\mathcal{L}_{\text{inter}} = \frac{1}{\bar{N} \binom{G}{2}} \sum_{i=1}^{\bar{N}} \sum_{g < g'} \mathrm{DKTau}\Big(\bar{\mathcal{S}}_i^{(g)},\ \bar{\mathcal{S}}_i^{(g')};\ \tau \Big). \tag{24}$$

The final ranking-aware regularizer is

$$\mathcal{L}_{\text{ratr}} = \lambda_{\text{intra}}\, \mathcal{L}_{\text{intra}} + \lambda_{\text{inter}}\, \mathcal{L}_{\text{inter}}. \tag{25}$$

Equations (23)–(25) play complementary roles. The intra term prevents two branches from assigning nearly identical orders to the most likely temporal correspondences, while the inter term prevents the same collapse on harder negatives. Minimizing both terms does

not force arbitrary disagreement because the restoration losses still anchor the recalibrated feature stream to the reconstruction target through MGFE.

3.7. Symmetry Property of the Regularizer

The representation symmetry introduced in Section 1 has a precise, operational meaning in MgfrOFR. It is not a geometric symmetry of the image content but an invariance of the diversity objective under the group actions that relabel or exchange restoration branches. We state this as two invariance properties, then explain why such invariance is a symmetry principle rather than a balancing heuristic, which is the distinction raised in the scope discussion of Section 2.5.

Property 1 (Branch-permutation invariance). Both the intra term in Equation (23) and the inter term $\mathcal{L}_{\text{inter}}$ sum the differentiable Kendall statistic over unordered branch pairs $g < g'$, and Equation (22) is symmetric in its two arguments, $\text{DKTau}(p, q; \tau) = \text{DKTau}(q, p; \tau)$. Applying the same permutation π to the branch index of every descriptor therefore leaves the regularizer unchanged:

$$\mathcal{L}_{\text{ratr}}\Big(\{b^{(g)}\}_{g=0}^{G-1}\Big) = \mathcal{L}_{\text{ratr}}\Big(\{b^{(\pi(g))}\}_{g=0}^{G-1}\Big), \qquad \forall\, \pi \in \mathcal{S}_G, \tag{26}$$

where $\mathcal{S}_G$ is the symmetric group on the G branches. Equivalently, $\mathcal{L}_{\text{ratr}}$ is a symmetric function of the branch set: it depends only on the unordered collection of pairwise branch relations, not on any branch ordering or index label. Because $\mathcal{S}_G$ has no privileged element, no single branch can be promoted to a dominant role without leaving the orbit of objective-preserving relabelings.

Property 2 (Direction-exchange invariance). The two propagation streams enter the regularizer only through the average $B_t = \frac{1}{2}(B_t^b + B_t^f)$ in Equation (2), which is invariant under the $\mathbb{Z}_2$ action that swaps the backward and forward descriptors, $B_t^b \leftrightarrow B_t^f$. The diversity objective therefore treats the two temporal propagation directions symmetrically and inherits the bidirectional symmetry of the underlying backbone.

Symmetry versus balancing. These two properties clarify why the proposed constraint is a symmetry rather than a tuned balancing term. Balancing describes a desired *outcome*—branches that end up contributing comparably—and is typically enforced by an asymmetric, hand-weighted penalty on the relative magnitude of individual branches. The RATR objective instead carries a structural symmetry: it is assembled from group-invariant quantities ($\mathcal{S}_G$ acting on branches and $\mathbb{Z}_2$ acting on directions) and applies an identical constraint to every branch by construction. Balanced and complementary branches are then a *consequence* of optimizing this symmetric objective, not a separately weighted goal. This is the precise sense in which branch balance is derived from a symmetry principle rather than imposed as one.

Scope of the claim. The symmetry asserted here is functional and representation-level: it is an invariance of the learning objective under permutation and exchange group actions, which are standard symmetry principles in machine learning, rather than a visible geometric symmetry of the restored frames. We state this restricted reading explicitly to avoid overclaiming, and Section 2.5 positions it against the broader use of symmetry in representation learning.

3.8. Training Objective

The generator objective combines the standard OFR losses used by the MambaOFR baseline with the proposed ranking-aware regularizer. Specifically, we use an L_1 pixel reconstruction loss $\mathcal{L}_{\text{pix}}$, whose robustness for image restoration objectives is analyzed in [36], a VGG-based perceptual loss $\mathcal{L}_{\text{perc}}$ [37], an adversarial loss $\mathcal{L}_{\text{gan}}$ [38], and the proposed ranking regularization $\mathcal{L}_{\text{ratr}}$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pix}} + \lambda_{\text{perc}}\mathcal{L}_{\text{perc}} + \lambda_{\text{gan}}\mathcal{L}_{\text{gan}} + \lambda_{\text{ratr}}\mathcal{L}_{\text{ratr}}. \tag{27}$$

Equation (27) preserves backward compatibility with the baseline training loop because MGFE only augments the propagated feature stream and RATR appears purely as an auxiliary regularizer. The loss weights are set to $\lambda_{\text{perc}} = 1.0$ and $\lambda_{\text{gan}} = 0.01$, following RTN [2] and MambaOFR [4]. For the ranking term, we use $\lambda_{\text{ratr}} = 0.1$, $\lambda_{\text{intra}} = 1.0$, and $\lambda_{\text{inter}} = 0.5$ (see Section 4.5 for sensitivity to these choices). The smaller weight on $\mathcal{L}_{\text{inter}}$ reflects the coarser ordering signal of clip-centroid negatives. Unless otherwise stated, the full model uses feature dimension $C_f = 16$, $G = 3$ branches, $M = 12$ restoration tokens, channel-wise average fusion in Equation (7), and a differentiable Kendall temperature of $\tau = 0.1$.

4. Experiment

4.1. Datasets and Evaluation Metrics

4.1.1. Datasets

We evaluate MgfrOFR on the SRWOV benchmark introduced with MambaOFR [4]. SRWOV is a dedicated old film restoration benchmark with both synthetic and real-world test sets.

Training uses the REDS training split [39], which contains 240 video sequences, each with 100 frames at 720×1280 resolution. Clean REDS frames are degraded on the fly using the synthetic degradation pipeline from MambaOFR. This pipeline simulates scratches, dust, blotches, flicker, and grain to produce paired degraded and clean training samples.

Evaluation uses two SRWOV test settings with different evidence types. The synthetic test set contains 30 clips rendered with the same degradation pipeline on held-out REDS sequences. Each clip consists of 100 frames at 720×1280 resolution and has paired ground truth, enabling full-reference metrics. The real-world set comprises 216 archival clips from the 1920s to 1960s with authentic mixed degradation and no ground truth; it is evaluated with no-reference quality metrics. Following the original SRWOV protocol [4], we use all 216 real-world clips, covering indoor portraits, newsreel footage, conversations, and multi-person scenes for a total of 29,256 frames.

4.1.2. Evaluation Metrics

We use seven metrics to cover pixel fidelity, perceptual similarity, and statistical naturalness. For synthetic evaluation with ground truth, PSNR and SSIM [40] measure pixel-level fidelity (higher is better), LPIPS [41] measures learned perceptual distance (lower is better), and DISTS [42] measures structure-and-texture perceptual similarity (lower is better). For real-world evaluation without ground truth, NIQE [43] and BRISQUE [44] measure statistical naturalness (lower is better), while CLIP-IQA+ [45] uses CLIP [46] visual features with learnable quality-aware prompts (higher is better). The metrics are reported separately when they emphasize different aspects of quality.

4.2. Implementation Details

4.2.1. Training Details

We train MgfrOFR on the REDS training split [39] with the synthetic degradation pipeline described in Section 4.1.1. Optimization uses Adam [47] with $\beta_1 = 0.9$ and $\beta_2 = 0.99$, an initial learning rate of 2×10^{-4}, and a MultiStep scheduler with milestones at 50K, 100K, and 150K iterations ($\gamma = 0.5$). Training runs for 200K iterations on 4 NVIDIA A100 GPUs (80 GB each), with one clip per GPU per mini-batch. Each training sample contains $T = 7$ consecutive frames cropped to 256×256. The software stack consists of PyTorch 2.7 and CUDA 12.4, with the random seed fixed to 42 for reproducibility. At

inference, RATR is disabled and frames are processed at full resolution (720×1280) without cropping.

4.2.2. Baseline Adaptation Protocol

The comparison uses released implementations and published settings, reproduced in a unified environment to keep the evaluation as fair as practical, and we make the per-baseline adaptation explicit along four axes.

Retraining and computational budget. Dedicated OFR methods (RTN, RRTN, MambaOFR) are retrained under the same REDS-based synthetic film degradation pipeline as MgfrOFR. In particular, MambaOFR—our primary baseline—and MgfrOFR are trained with an identical 200K-iteration schedule on the same A100-class hardware, so their computational budget for OFR adaptation is matched; RTN and RRTN are reproduced under the same degradation pipeline with a comparable training budget.

Hyper-parameter tuning. We retain each baseline's published architecture-specific hyper-parameters (optimizer and learning-rate schedule) rather than performing exhaustive per-baseline re-tuning. Re-tuning every method to convergence on OFR would be computationally prohibitive and could itself bias the comparison toward whichever baseline received the most tuning effort.

Degradation adaptation. Generic video restoration methods (VRT, RVRT, ShiftNet) keep their published architecture and default optimization recipe; we replace only the task-specific degradation generation so that they can be trained on OFR-style inputs, and we report them as faithfully reproduced references whose training budget follows each released recipe.

Reference systems. Enhancement-oriented methods (DeOldify, OldPhoto, DeepRemaster) are not designed for paired old film restoration and are not retrained on the OFR pipeline; we include them strictly as reference systems to contextualize perceptual trade-offs, not as task-matched OFR baselines.

We therefore interpret the comparison as a unified reproduction study under a common degradation protocol with a budget-matched dedicated-OFR tier, not as a claim that every baseline has been exhaustively re-optimized for OFR.

4.3. Comparison with State-of-the-Art

We compare MgfrOFR against nine baselines spanning dedicated OFR methods, general video restoration models, and image/video enhancement approaches. The dedicated OFR baselines are RTN [2], RRTN [3], and MambaOFR [4]; the general video restoration baselines are VRT [18], RVRT [19], and ShiftNet [48]; and the enhancement-oriented references are DeOldify [49], OldPhoto [50], and DeepRemaster [1]. No-reference metrics are computed using the pyiqa library under the same software environment.

4.3.1. Quantitative Comparison on Synthetic Data

Table 2 reports full-reference evaluation on the SRWOV synthetic benchmark. MgfrOFR achieves the best PSNR/SSIM and the lowest LPIPS (0.0600) and DISTS (0.0656), surpassing the strongest baseline MambaOFR by +1.02 dB PSNR under the same 30-clip REDS split protocol. The pattern is consistent with the intended architectural distinction: generic video restoration methods do not include OFR-specific defect handling, while frame-oriented enhancement approaches can suffer from temporal inconsistency.

To test whether the gain over MambaOFR is clip-consistent, we apply paired, two-sided Wilcoxon signed-rank tests to the 30 per-clip means under the same REDS synthetic protocol. MgfrOFR improves PSNR on 29/30 clips ($p = 3.73 \times 10^{-9}$; median ΔPSNR $= 0.686$ dB, IQR [0.597, 1.176] dB) and SSIM on 30/30 clips ($p = 1.73 \times 10^{-6}$; median ΔSSIM $= 0.0091$). The perceptual full-reference metrics show the same clip-level direction:

Table 2. Quantitative comparison reports SRWOV synthetic full-reference results under the unified reproduction protocol. Best results are in **bold**, second best are underlined.

Method	PSNR↑	SSIM↑	LPIPS↓	DISTS↓
ShiftNet [48]	21.48	0.8234	0.2030	0.1303
VRT [18]	22.39	0.8631	0.1750	0.1087
RVRT [19]	22.43	0.8710	0.1655	0.1078
DeOldify [49]	23.22	0.8923	0.2775	0.1260
RRTN [3]	24.15	0.9006	0.1082	0.0668
OldPhoto [50]	24.62	0.8524	0.2252	0.1144
DeepRemaster [1]	25.06	0.9185	0.1108	0.0895
RTN [2]	25.85	0.8869	0.1214	0.0862
MambaOFR [4]	26.84	0.9153	0.0719	0.0733
MgfrOFR (Ours)	**27.86**	**0.9255**	**0.0600**	**0.0656**

LPIPS improves on 30/30 clips ($p = 1.86 \times 10^{-9}$; median $\Delta = -0.0113$) and DISTS improves on 29/30 clips ($p = 1.92 \times 10^{-6}$; median $\Delta = -0.0074$). The matched-pairs rank-biserial correlations are 0.996, 1.000, 1.000, and 0.996 for PSNR, SSIM, LPIPS, and DISTS, respectively. These statistics support the F1 main-model choice while keeping the real-world no-reference metrics interpreted separately.

4.3.2. Quantitative Comparison on Real-World Data

Table 3 reports no-reference quality metrics on 216 real-world archival clips from the SRWOV real-world set, where ground truth is unavailable. Because NIQE and BRISQUE emphasize statistical naturalness whereas CLIP-IQA+ reflects CLIP-aligned perceptual quality, we report the metrics separately rather than collapsing them into a single aggregate ranking. All methods are evaluated on the full 216-clip set under the same pyiqa-based metric environment.

Table 3. Real-world evaluation reports no-reference quality metrics on SRWOV real footage under the same pyiqa-based metric environment.

Method	NIQE↓	BRISQUE↓	CLIP-IQA+↑
ShiftNet [48]	5.71	42.82	0.3975
VRT [18]	5.66	42.73	0.3985
RVRT [19]	5.72	42.64	0.4013
DeOldify [49]	5.72	41.61	0.3768
RRTN [3]	5.67	32.86	0.3969
OldPhoto [50]	6.56	30.45	**0.4503**
DeepRemaster [1]	5.99	40.62	0.3991
RTN [2]	5.17	19.00	0.4163
MambaOFR [4]	5.34	29.79	0.4351
MgfrOFR (Ours)	**5.12**	24.79	0.4092

MgfrOFR obtains the best NIQE (5.12) among the compared methods and improves BRISQUE relative to MambaOFR (24.79 vs. 29.79), though it remains behind RTN (19.00) on BRISQUE. The dataset average improves by -4.0% on NIQE relative to the previous best RTN (5.17), indicating better statistical naturalness under this no-reference metric. On CLIP-IQA+, MgfrOFR (0.4092) ranks below both OldPhoto (0.4503) and MambaOFR (0.4351); this pattern reflects that CLIP-IQA+ favors the sharper, enhancement-oriented restoration style of OldPhoto, whereas our conservative granularity recalibration prioritizes statistical naturalness. Because the three no-reference metrics quantify different perceptual attributes, these results should not be collapsed into a single superiority claim. The trade-off between naturalness and enhancement-oriented perceptual quality is discussed in Section 4.7.4.

4.3.3. Qualitative Comparison

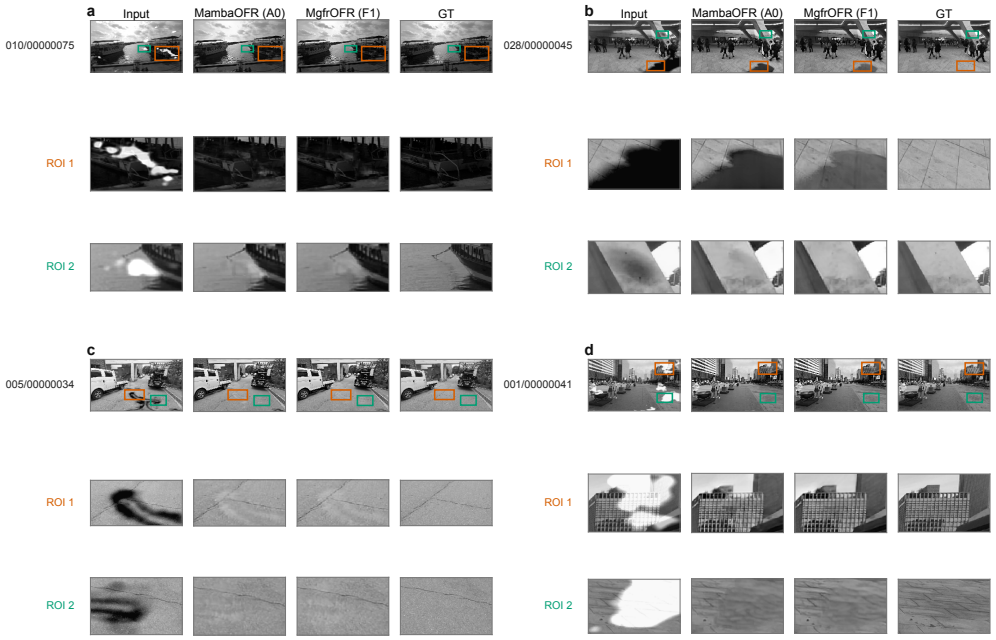


Figure 4. Focused high-resolution ROI comparison on four representative SRWOV synthetic scenes. Each scene is shown with the degraded input, MambaOFR baseline, MgfrOFR(F1), and the ground truth. Red and green boxes mark local regions in each overview row, and the enlarged ROI tiles below each row expose scratch removal, local edge preservation, and texture recovery. The comparison illustrates that MgfrOFR(F1) more closely follows the ground-truth structures while reducing visible old-film artifacts across the selected scenes.

Figure 4 compares focused ROI results on four representative synthetic scenes from the SRWOV benchmark. The layout uses a compact input–MambaOFR–MgfrOFR(F1)–GT comparison so that each ROI can be enlarged without compressing the local evidence. In these ROI tiles, MgfrOFR(F1) more closely follows the ground-truth edge layout and local texture around scratches, blotches, and high-contrast damaged regions while suppressing visible old-film artifacts. These visual observations are consistent with the PSNR advantage reported in Table 2.

Figure 5 compares visual results on four representative clips from the SRWOV real-world set: FV2SEyszKT0/156–161/00015, FV2SEyszKT0/906–911/00077, 6ZCNaCHPacE/236–241/00013, and gRaosDFqTfo/395–400/00080. Because no ground truth is available, the comparison focuses on visible artifact suppression, structural preservation, and naturalness. Compared with MambaOFR, MgfrOFR(F1) more consistently preserves facial structure, fabric texture, object boundaries, and local contrast while suppressing residual old-film artifacts in these examples. Supplementary Video S1 provides the corresponding side-by-side temporal comparison on three real-world clips. These qualitative observations align with MgfrOFR's best NIQE and improved BRISQUE relative to MambaOFR in Table 3; the CLIP-IQA+ ranking remains reported separately because it favors OldPhoto in the aggregate metric. MgfrOFR may still over-smooth very fine textures under extreme degradation, and severe motion blur can still cause flow-alignment errors; these cases are discussed in Section 4.7.4.

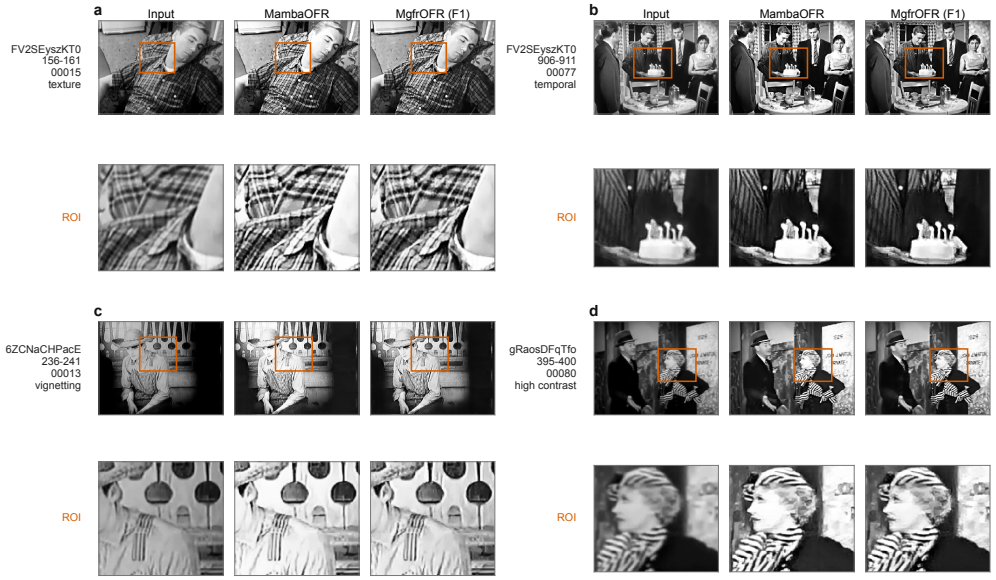


Figure 5. Focused high-resolution ROI comparison on four representative SRWOV real-world clips. Each scene is shown with the degraded input, MambaOFR, and MgfrOFR(F1), with annotated ROI tiles targeting texture recovery, temporal artifact suppression, vignetting, and high-contrast boundaries. Because real-world clips lack ground truth, the comparison focuses on visible artifact suppression, structural preservation, and naturalness.

4.4. Ablation Study

Table 4 separates the effects of MGFE recalibration and RATR regularization. The setting names are kept consistent with Figure 6. The +RATR only setting is a partial architectural control: MGFE branch descriptors are still computed as RATR ranking targets, while channel-wise recalibration is disabled.

Table 4. Ablation study on the SRWOV synthetic benchmark. The setting names match Figure 6; "MGFE" denotes channel-wise recalibration in this table. Best results are in **bold**.

Setting	MGFE	RATR	PSNR↑	SSIM↑	LPIPS↓	DISTS↓
MambaOFR (A0)	Off	Off	26.84	0.9153	0.0719	0.0733
+MGFE only (A1)	On	Off	27.75	0.9219	0.0651	0.0702
+RATR only (A2)	Off	On	27.72	0.9246	0.0646	0.0696
MgfrOFR (F1, Ours)	On	On	**27.86**	**0.9255**	**0.0600**	**0.0656**
+ max fusion (A3)	On	On	27.17	0.9168	0.0740	0.0773

Discussion. MgfrOFR (F1) improves MambaOFR by +1.02 dB in PSNR and achieves the best results across all four metrics. +MGFE only (A1) improves the baseline but remains weaker than F1 on LPIPS and DISTS, showing that recalibration alone is not sufficient for the best fidelity–perception balance. +RATR only (A2) gives a comparable PSNR gain and better perceptual balance than A1, confirming that ranking diversity already helps before channel-wise feedback is enabled. The full F1 model with average fusion gives the strongest combined result. The max-fusion configuration (A3), retained as a fusion-ablation control, shows negative synergy on all four full-reference metrics, justifying the switch from max to average fusion as the default.

4.5. Hyperparameter Sensitivity

We analyze only sensitivity entries that are backed by frozen test logs. Because the main model changed from max fusion to average fusion after model selection, the old pre-D2 max-fusion sweep is not reused as F1 evidence. Table 5 therefore reports the measured granularity controls and the measured lower-temperature ranking control, while

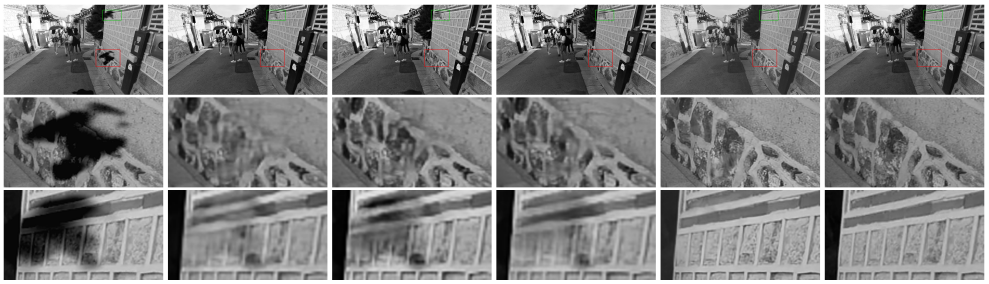


Figure 6. Visual ablation on REDS-based SRWOV synthetic data. **Top row:** full frames with red and green crop boxes marking the zoom regions. **Middle and bottom rows:** zoomed patches from the marked regions. From left to right, the restoration columns compare MambaOFR (A0), +MGFE only (A1), +RATR only (A2), and MgfrOFR (F1, Ours), with the degraded input and ground truth included as references. The max-fusion ablation (A3) is omitted for visual clarity. The comparison demonstrates that combining MGFE with RATR under average fusion yields the cleanest local restoration among the ablated configurations in Table 4.

unmeasured token-insertion and aggregation variants are excluded from the manuscript conclusions.

Table 5. Measured sensitivity controls on REDS synthetic data. F1 is the average-fusion main model. G1/G2/G4 are granularity controls trained with max fusion, and R1 is a lower-temperature ranking control; these controls are used to assess robustness, not to replace the F1 main-model selection.

Variant	Tokens	Branches	PSNR↑	SSIM↑	LPIPS↓	DISTS↓
G1 (4t/1b)	4	1	**27.95**	**0.9263**	0.2153	0.1554
G2 (8t/2b)	8	2	27.49	0.9211	0.2162	0.1557
F1 (12t/3b, avg)	12	3	27.86	0.9255	**0.0600**	**0.0656**
G4 (16t/4b)	16	4	27.53	0.9197	0.2169	0.1565
R1 (τ=0.05)	12	3	26.87	0.9177	0.0672	0.0717

The granularity controls show a clear fidelity–perception split. G1 obtains the highest PSNR/SSIM but much worse LPIPS/DISTS, indicating that the single-branch setting overfits pixel fidelity and loses perceptual balance. F1 is not the absolute PSNR maximum, but it is the only measured setting in this group that combines near-best fidelity with the best LPIPS and DISTS. The lower-temperature R1 control remains competitive on LPIPS/DISTS but loses fidelity relative to F1, so $\tau = 0.1$ is retained for the main model.

4.6. Complexity Analysis

Table 6 reports computational complexity for seven-frame 256×256 clips. To keep the comparison auditable, the four baseline rows reuse the frozen 007 round-2 L40 unified benchmark artifacts generated under the same executable BasicOFR protocol (FP32, batch size 1, three warm-up and ten timed runs). The proposed F1 row is newly verified for this revision on the L40 server with the current 001 configuration. FLOPs and latency are reported per seven-frame clip; the flow-based methods include the executable optical-flow front end, while DeepRemaster uses no flow estimator.

The matched 001 A0/F1 sanity benchmark confirms that the F1 configuration adds 0.012M parameters over the project MambaOFR baseline (8.754M vs. 8.742M) and 14.094G FLOPs per seven-frame clip (3107.781G vs. 3093.687G). The table therefore supports two claims separately: the baseline methods are compared using the reused frozen 007 L40 runtime artifacts, and the current 001 F1 configuration has newly verified L40 runtime, FPS, and peak-memory evidence for this revision.

Table 6. Computational complexity comparison under the L40 FP32 unified benchmark protocol (1×7×3×256×256, batch size 1, three warm-up and ten timed runs). The DeepRemaster/RTN/RRTN/MambaOFR baseline rows reuse the 007 round-2 L40 unified benchmark artifacts under the same executable BasicOFR protocol; the MgfrOFR(F1) row was newly verified for this revision with the current 001 configuration. Latency is reported per seven-frame clip.

Method	Params (M)	FLOPs (G)	Latency (ms/clip)↓	FPS↑	Peak Mem. (GB)↓
DeepRemaster [1]	9.8781	914.0	17.9	391.6	0.35
RTN [2]	5.7351	2184.7	169.5	41.3	0.24
RRTN [3]	7.0503	2782.2	298.4	23.5	0.75
MambaOFR [4]	8.2984	2832.8	404.8	17.3	0.60
MgfrOFR(F1 Ours)	8.7540	3107.8	488.5	14.3	0.61

4.7. Analysis and Limitations

4.7.1. Comparison with the Closest Prior Work

MgfrOFR is built directly on top of MambaOFR [4], so its contribution should be distinguished from the baseline backbone. We do not claim originality for bidirectional Mamba propagation, the degradation-aware prompt block, or the original reconstruction head. The contribution of this work is the reorganization and regularization of propagated features before reconstruction.

Compared with MambaOFR, MgfrOFR differs in four implementation-level aspects. First, the propagated feature is no longer treated as a single restoration carrier; MGFE exposes explicit coarse-to-fine branches. Second, these branches are summarized and fed back through channel-wise recalibration rather than used only for auxiliary supervision. Third, branch diversity is constrained by ranking correlation on temporal correspondences rather than by an identity-based objective from recognition tasks. Fourth, RATR is used only during training, so the deployment interface remains compatible with the baseline inference pipeline.

The symmetry-related claim is limited to representation balance. MgfrOFR does not model object-level geometric regularity and does not propose a general mathematical theory of symmetry. Instead, it studies whether coarse-to-fine restoration branches can remain balanced under asymmetric film degradation. MGFE separates evidence by restoration granularity, and RATR discourages redundant temporal-similarity rankings across branches.

4.7.2. MGFE–RATR Synergy for Branch Balance

The ablation study indicates that MGFE and RATR have different roles in the full (F1) model. RATR acts as a diversity regularizer that makes branches more complementary, while MGFE recalibration alone mainly improves fidelity (Table 4). In the RATR-only configuration, MGFE branch descriptors are still computed and used as multi-granularity ranking targets; only the channel-wise scale/shift feedback is disabled. MGFE therefore creates the branch structure needed for coarse-to-fine feature balance, whereas RATR suppresses redundant temporal-ranking behavior.

The asymmetric ablation behavior in Table 4 shows that MGFE recalibration alone improves the baseline but does not reach the fidelity–perception balance of the full F1 model. This is consistent with the perception–distortion trade-off [51], which establishes that optimizing a distortion measure such as PSNR does not by itself guarantee the best perceptual behavior. The channel-wise recalibration in Equation (14) is conditioned on an averaged branch context and is driven mainly by the pixel reconstruction loss. Without a

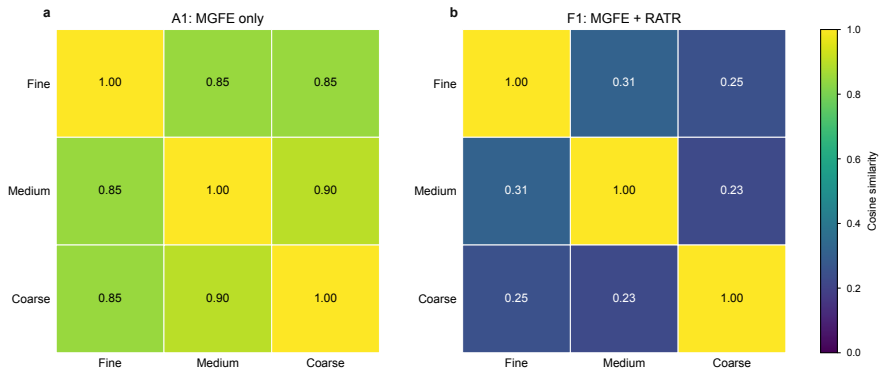


Figure 7. Branch balance analysis. Pairwise cosine similarity is compared between the MGFE-only configuration and the full (F1) MgfrOFR model under the 30-clip REDS synthetic protocol. RATR reduces off-diagonal similarity from 0.85–0.90 to 0.23–0.31, confirming that ranking-aware regularization prevents granularity branches from collapsing into correlated descriptors.

diversity constraint, the granularity branches can converge toward similar low-frequency corrections, so the MGFE-only variant remains less effective than F1 on LPIPS and DISTS. RATR mitigates this effect by acting on branch-ranking structure rather than on pixel error. Penalizing correlated temporal rankings across branches (Equations (23)–(25)) prevents the branches from collapsing onto a single fidelity-oriented direction and preserves complementary fine-grained evidence. The full (F1) model therefore gives the strongest combined fidelity and perceptual result.

The real-world results are consistent with this interpretation but do not prove content fidelity without ground truth. On the SRWOV real-world test set (Table 3), MgfrOFR achieves the best NIQE (5.12) among the compared methods and improves BRISQUE relative to MambaOFR (24.79 vs. 29.79), while its CLIP-IQA+ (0.4092) is lower than both OldPhoto and MambaOFR. This pattern suggests that the multi-granularity decomposition transfers useful restoration structure beyond the synthetic training distribution, but the no-reference gains should be interpreted as evidence of statistical naturalness rather than definitive perceptual superiority.

4.7.3. Branch Balance Analysis

The branch analysis assesses whether the granularity branches remain complementary rather than becoming correlated copies of the same descriptor. We compute pairwise cosine similarity between branch descriptors on the SRWOV synthetic test set. This analysis directly supports the representation-balance claim: useful branches should remain comparable but not redundant.

The full (F1) model yields lower inter-branch similarity than the MGFE-only variant (Figures 7–9). This supports the role of RATR as the mechanism that keeps granularity branches functionally distinct. The RATR-only variant remains closer to the full (F1) model than to MGFE-only, which further indicates that ranking diversity regularization suppresses collapse even before recalibration feedback is enabled.

Figure 7 illustrates this contrast for a representative frame. In the MGFE-only configuration, the off-diagonal similarities remain high, indicating that the branches encode nearly the same restoration evidence. In the full (F1) model, these off-diagonal values decrease markedly, especially for the fine-grained branch, showing a more differentiated branch geometry.

The inspected examples suggest a consistent specialization pattern. The fine-grained branch is most sensitive to localized defects such as scratches and pixel-level noise; the coarse branch emphasizes broader structural corruption such as blotches and cracks; and the medium branch occupies an intermediate role. This behavior is consistent with progres-

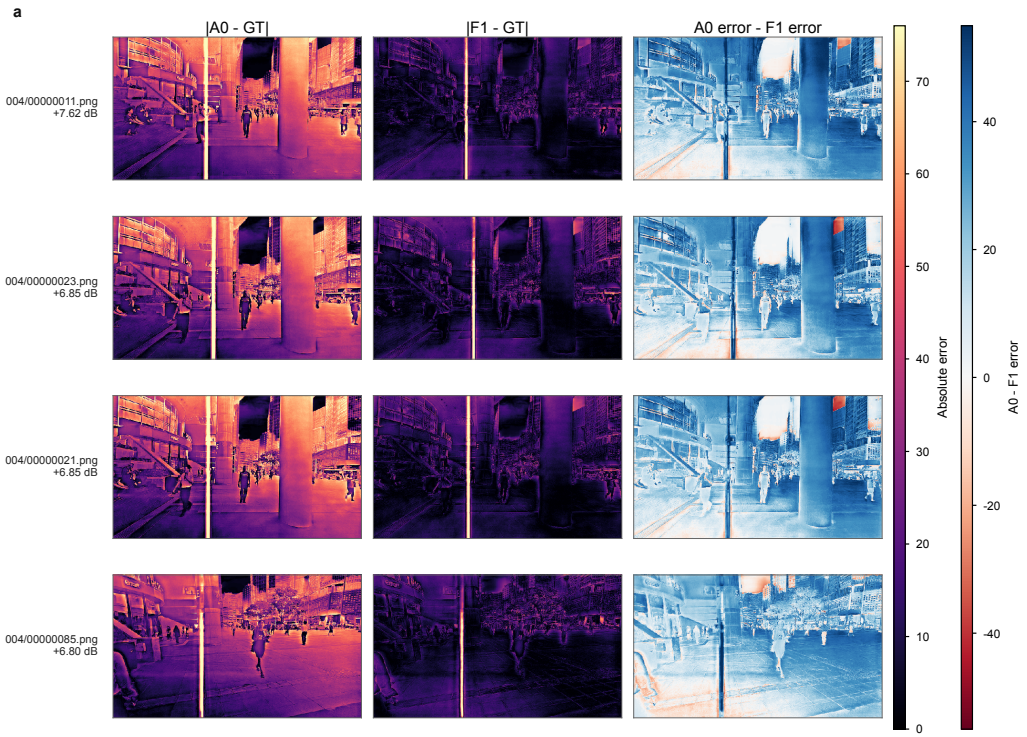


Figure 8. Pixel-wise absolute-error comparison on four gate-selected synthetic frames. The left and middle columns compare $|A0 - GT|$ and $|F1 - GT|$; the right column illustrates their difference, where red indicates regions in which F1 reduces the reconstruction error. The error maps demonstrate that lower branch correlation translates into spatially broader restoration gains rather than isolated local fixes.

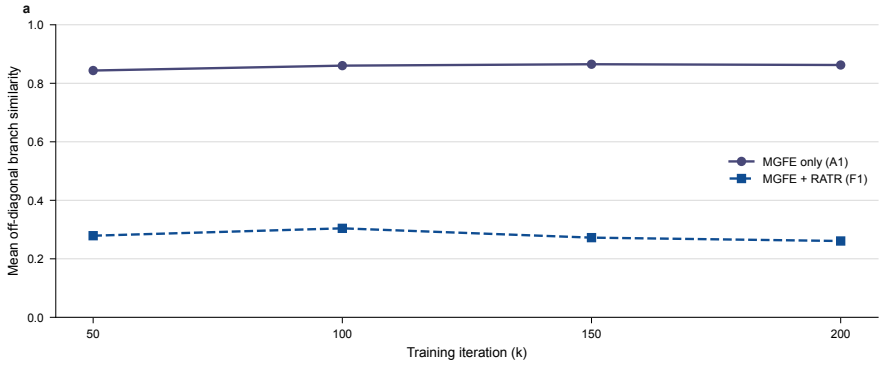


Figure 9. Training-time evolution of mean off-diagonal branch similarity. The curves compare A1 (MGFE only) and F1 (MGFE+RATR, average fusion) at 50k, 100k, 150k, and 200k iterations on the same 30-clip REDS synthetic protocol. The F1 curve remains lower after RATR is introduced, supporting the claim that ranking-aware regularization keeps granularity branches complementary during training rather than only at the final checkpoint.

sive token fusion, where fewer surviving tokens aggregate over wider spatial support. The pixel-domain error maps in Figure 8 further indicate that branch decorrelation corresponds to broader restoration gains rather than isolated local fixes, while Figure 9 shows that this separation is sustained during training.

4.7.4. When Does MgfrOFR Struggle?

MgfrOFR's advantage is not uniform across all degradation conditions. The per-clip ranking shows that F1 is weakest relative to A0 on clips 029, 022, and 017, suggesting that the full model is most useful when heterogeneous degradation is present, such as structural damage co-occurring with fine scratches in the same frame. On clips dominated by a single

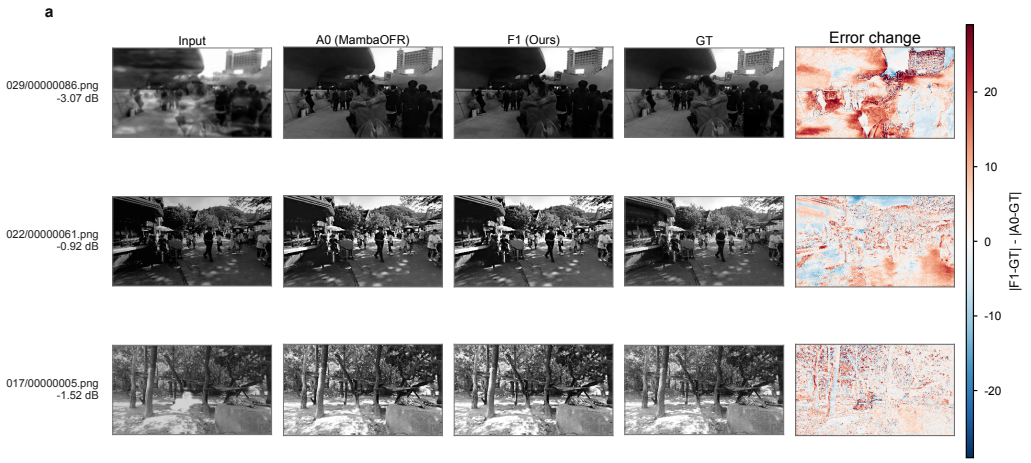


Figure 10. Least-improved synthetic cases selected from the frozen A0-versus-F1 per-frame ranking within clips 029, 022, and 017. Columns show the degraded input, A0, F1, GT, and the signed error-difference map $|F1–GT| - |A0–GT|$, where red marks regions in which F1 has larger error than A0. These examples clarify that MgfrOFR(F1) can be less favorable when the degradation is dominated by a narrow or weakly heterogeneous corruption pattern.

degradation type, such as uniform flicker without spatial damage, the multi-granularity decomposition can add representational overhead without proportional benefit, and the baseline or RATR-only variant may match the full (F1) model's fidelity.

The real-world CLIP-IQA+ result highlights a trade-off between naturalness and enhancement-oriented perceptual quality. MgfrOFR's CLIP-IQA+ (0.4092) is lower than both OldPhoto (0.4503) and MambaOFR (0.4351), while MgfrOFR achieves the best NIQE (5.12) among the compared methods and improves BRISQUE relative to MambaOFR (24.79 vs. 29.79). This pattern reflects that different no-reference metrics measure distinct aspects of restoration quality. CLIP-IQA+ is trained on human quality annotations that correlate with sharp textures and high-contrast details [45], favoring methods such as OldPhoto that enhance perceptual sharpness through learned global priors. NIQE and BRISQUE instead measure statistical deviation from natural image distributions and penalize artifacts introduced by texture hallucination. MgfrOFR's granularity-based recalibration operates conservatively on the propagated feature stream, which appears to preserve statistical naturalness while providing less texture enhancement than OldPhoto. For archival restoration, NIQE/BRISQUE may be more relevant proxies when content fidelity is prioritized, but a human study is needed to validate that preference.

4.7.5. Limitations, Threats to Validity, and Future Work

Threats to Validity.

Five threats affect the interpretation of the results. **Dataset threat**: all training uses synthetic degradation applied to REDS clips, which introduces a synthetic-to-real domain gap and may limit generalization to archival footage with correlated degradation chains not modeled by the pipeline. **Metric threat**: evaluation relies on PSNR/SSIM/LPIPS/DISTS for synthetic full-reference assessment and no-reference metrics for real-world assessment; these metrics emphasize different aspects of quality and may not fully align with human perceptual judgments. **Human-evaluation threat**: the study does not include an expert reader study with film-restoration practitioners or archivists, so the practical value of improvements on authentic footage is inferred from proxy metrics and qualitative inspection. **Baseline threat**: three baselines (VRT, RVRT, ShiftNet) required adaptation from video restoration to OFR-specific degradation, which preserves their architectures but may not reflect their optimal performance under a different training regime. **Synthetic-to-real threat**: the real-world evaluation lacks ground truth, so improvements in NIQE/BRISQUE

reflect statistical naturalness rather than verified content fidelity, and the absence of external validation on additional datasets weakens the generalization claim.

Limitations and Future Work.

RATR adds a training-time cost because it requires branch-wise similarity matrices over the batch-time descriptor set $\bar{N} = BT$. Its dominant cost therefore grows approximately quadratically in $\bar{N}$. In our setting (B=4, T=7, G=3), the total overhead of MGFE and RATR is 0.5 GB (7.0 vs. 6.5 GB; Table 6); however, longer clips or larger effective batch-time products would make this quadratic term more pronounced. Gradient checkpointing or selective ranking construction, such as sampling a subset of anchors per iteration, will be needed for larger training settings. The second limitation is the synthetic-to-real domain gap caused by REDS-based training degradation. Although the real-world NIQE/BRISQUE results are encouraging, authentic archival footage can contain correlated degradation chains that synthetic pipelines do not model. The third limitation is evaluation scope: validation is limited to SRWOV, and additional archival datasets plus human evaluation would strengthen the generalization claim.

Promising extensions include adaptive branch activation conditioned on degradation severity, frequency-aware branch specialization that assigns structural recovery and texture restoration to dedicated branches, and joint OFR–colorization frameworks for archival digitization and cultural heritage restoration workflows [12].

5. Conclusions

We presented MgfrOFR, a Mamba-based old film restoration framework that augments MambaOFR with explicit multi-granularity feature balancing. MGFE reorganizes propagated features into coarse-to-fine restoration-token branches and recalibrates the spatial stream through channel-wise feedback. RATR discourages branch collapse by penalizing correlated temporal-similarity rankings during training while preserving the baseline inference interface.

The SRWOV results support the claim that coarse-to-fine branch balance improves old film restoration under heterogeneous degradation. Under the unified reproduction protocol on the SRWOV synthetic benchmark, MgfrOFR achieves the strongest full-reference scores among the tested baselines, improving PSNR by 1.02 dB over MambaOFR (27.86 dB vs. 26.84 dB) with only 0.14% additional parameters. On 216 real-world archival clips, MgfrOFR obtains the lowest NIQE (5.12) among the compared methods and improves BRISQUE relative to MambaOFR (24.79 vs. 29.79), while its CLIP-IQA+ (0.4092) remains below OldPhoto and MambaOFR, reflecting a conservative restoration style that favors statistical naturalness over texture enhancement. Because the real-world setting lacks ground truth, these no-reference gains should be interpreted as evidence of improved statistical naturalness rather than definitive proof of perceptual superiority. The ablation study further shows that MGFE and RATR are complementary: MGFE alone improves the baseline but remains weaker than F1 on the perceptual full-reference metrics, whereas the full model gives the best combined result when branch diversity is explicitly enforced through RATR and average fusion.

Several limitations define the scope of the conclusion. The method inherits the synthetic-to-real domain gap of REDS-based training; RATR adds a training-time cost that grows approximately quadratically with the batch-time descriptor set and increases training time by approximately 7.6% over MambaOFR under our setting; and evaluation is limited to the SRWOV benchmark plus no-reference proxies on authentic footage. Future work will focus on adaptive branch activation conditioned on degradation severity, frequency-aware branch specialization that assigns structural recovery and texture restora-

tion to dedicated branches, lighter ranking constructions such as selective anchor sampling, human-in-the-loop evaluation, and broader validation on additional archival benchmarks.

Author Contributions: Conceptualization, F.C. and Y.D.; methodology, F.C.; software, F.C.; validation, F.C. and Q.Z.; formal analysis, F.C.; investigation, F.C.; resources, K.Y., L.C. and Y.D.; data curation, F.C.; writing—original draft preparation, F.C.; writing—review and editing, Q.Z., K.Y., L.C. and Y.D.; visualization, F.C.; supervision, Y.D.; project administration, Y.D.; funding acquisition, Y.D. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Supplementary Materials: Supplementary Video S1 provides side-by-side temporal comparisons of the degraded input, MambaOFR baseline, and MgfrOFR(F1) on representative SRWOV real-world clips.

Data Availability Statement: The training data follow the public REDS benchmark and the SRWOV protocol described by MambaOFR. The manuscript source and figures are included in this submission package. Code, configurations, and evaluation scripts for this study are publicly available at https://github.com/wscffaa/MgfrOFR. Public benchmark data should be obtained from the original REDS and SRWOV/MambaOFR sources; no new dataset is redistributed in this work.

Acknowledgments: The authors thank the developers of MambaOFR for releasing their code and pretrained models.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

OFR	Old Film Restoration
MgfrOFR	Proposed old film restoration framework in this work
MGFE	Multi-Granularity Feature Extractor
RATR	Ranking-Aware Triplet Regularization
SSM	State Space Model
PSNR	Peak Signal-to-Noise Ratio
SSIM	Structural Similarity Index
LPIPS	Learned Perceptual Image Patch Similarity
DISTS	Deep Image Structure and Texture Similarity
NIQE	Natural Image Quality Evaluator
BRISQUE	Blind/Referenceless Image Spatial Quality Evaluator
CLIP-IQA+	CLIP-based Image Quality Assessment (enhanced)
SRWOV	Synthetic + Real-World Old Video
REDS	REalistic and Diverse Scenes
GT	Ground Truth
CNN	Convolutional Neural Network
GPU	Graphics Processing Unit

References

1. Iizuka, S.; Simo-Serra, E. DeepRemaster: Temporal Source-Reference Attention Networks for Comprehensive Video Enhancement. *ACM Trans. Graph.* **2019**, *38*, 1–13. https://doi.org/10.1145/3355089.3356570.
2. Wan, Z.; Zhang, B.; Chen, D.; Liao, J. Bringing Old Films Back to Life. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, LA, USA, 19–24 June 2022; pp. 17694–17703. https://doi.org/10.1109/CVPR52688.2022.01717.
3. Lin, S.; Simo-Serra, E. Restoring Degraded Old Films with Recursive Recurrent Transformer Networks. In Proceedings of the Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Waikoloa, HI, USA, 3–8 January 2024; pp. 6704–6714. https://doi.org/10.1109/WACV57701.2024.00658.

4. Mao, Y.; Luo, H.; Zhong, Z.; Chen, P.; Zhang, Z.; Wang, S. Making Old Film Great Again: Degradation-Aware State Space Model for Old Film Restoration. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA, 10–17 June 2025; pp. 28039–28049. https://doi.org/10.1109/CVPR52734.2025.02611.
5. Gu, A.; Dao, T. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. In Proceedings of the The First Conference on Language Modeling (COLM), Philadelphia, PA, USA, 7–10 October 2024.
6. Guo, H.; Li, J.; Dai, T.; Ouyang, Z.; Ren, X.; Xia, S.T. MambaIR: A Simple Baseline for Image Restoration with State-Space Model. In Proceedings of the Computer Vision – ECCV 2024, Milan, Italy, 29 September–4 October 2024; pp. 222–241. https://doi.org/10.1007/978-3-031-72649-1_13.
7. Lin, T.Y.; Dollár, P.; Girshick, R.; He, K.; Hariharan, B.; Belongie, S. Feature Pyramid Networks for Object Detection. In Proceedings of the Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA, 21–26 July 2017; pp. 936–944. https://doi.org/10.1109/CVPR.2017.106.
8. Wang, J.; Sun, K.; Cheng, T.; Jiang, B.; Deng, C.; Zhao, Y.; Liu, D.; Mu, Y.; Tan, M.; Wang, X.; et al. Deep High-Resolution Representation Learning for Visual Recognition. *IEEE Trans. Pattern Anal. Mach. Intell.* **2021**, *43*, 3349–3364. https://doi.org/10.1109/TPAMI.2020.2983686.
9. Sun, Y.; Zheng, L.; Yang, Y.; Tian, Q.; Wang, S. Beyond Part Models: Person Retrieval with Refined Part Pooling (and A Strong Convolutional Baseline). In Proceedings of the Computer Vision – ECCV 2018, Munich, Germany, 8–14 September 2018; pp. 501–518. https://doi.org/10.1007/978-3-030-01225-0_30.
10. Wang, G.; Yuan, Y.; Chen, X.; Li, J.; Zhou, X. Learning Discriminative Features with Multiple Granularities for Person Re-Identification. In Proceedings of the Proceedings of the ACM International Conference on Multimedia (MM), Seoul, South Korea, 22–26 October 2018; pp. 274–282. https://doi.org/10.1145/3240508.3240552.
11. Gu, H.; Yang, Q.; Pu, L.; Han, S.; Ding, Y. ReIDMamba: Learning Discriminative Features with Visual State Space Model for Person Re-Identification. *IEEE Trans. Multimedia* **2026**, pp. 1–12. Early Access, https://doi.org/10.1109/TMM.2026.3668499.
12. Kokaram, A.C. On Missing Data Treatment for Degraded Video and Film Archives: A Survey and a New Bayesian Approach. *IEEE Trans. Image Process.* **2004**, *13*, 397–415. https://doi.org/10.1109/TIP.2004.823815.
13. Dabov, K.; Foi, A.; Katkovnik, V.; Egiazarian, K. Image Denoising by Sparse 3-D Transform-Domain Collaborative Filtering. *IEEE Trans. Image Process.* **2007**, *16*, 2080–2095. https://doi.org/10.1109/TIP.2007.901238.
14. He, K.; Sun, J.; Tang, X. Guided Image Filtering. *IEEE Trans. Pattern Anal. Mach. Intell.* **2013**, *35*, 1397–1409. https://doi.org/10.1109/TPAMI.2012.213.
15. Wang, X.; Chan, K.C.K.; Yu, K.; Dong, C.; Loy, C.C. EDVR: Video Restoration with Enhanced Deformable Convolutional Networks. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Long Beach, CA, USA, 16–17 June 2019; pp. 1954–1963. https://doi.org/10.1109/CVPRW.2019.00247.
16. Chan, K.C.K.; Wang, X.; Yu, K.; Dong, C.; Loy, C.C. BasicVSR: The Search for Essential Components in Video Super-Resolution and Beyond. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA, 19–25 June 2021; pp. 4947–4956. https://doi.org/10.1109/CVPR46437.2021.00491.
17. Chan, K.C.K.; Zhou, S.; Xu, X.; Loy, C.C. BasicVSR++: Improving Video Super-Resolution with Enhanced Propagation and Alignment. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, LA, USA, 19–24 June 2022; pp. 5972–5981. https://doi.org/10.1109/CVPR52688.2022.00588.
18. Liang, J.; Cao, J.; Fan, Y.; Zhang, K.; Ranjan, R.; Li, Y.; Timofte, R.; Van Gool, L. VRT: A Video Restoration Transformer. *IEEE Trans. Image Process.* **2024**, *33*, 2171–2182. https://doi.org/10.1109/TIP.2024.3372454.
19. Liang, J.; Fan, Y.; Xiang, X.; Ranjan, R.; Ilg, E.; Green, S.; Cao, J.; Zhang, K.; Timofte, R.; Van Gool, L. Recurrent Video Restoration Transformer with Guided Deformable Attention. In Proceedings of the Advances in Neural Information Processing Systems, New Orleans, LA, USA, 28 November–9 December 2022; Vol. 35, pp. 378–393. https://doi.org/10.52202/068431-0028.
20. Lin, X.; He, J.; Chen, Z.; Lyu, Z.; Dai, B.; Yu, F.; Qiao, Y.; Ouyang, W.; Dong, C. DiffBIR: Toward Blind Image Restoration with Generative Diffusion Prior. In Proceedings of the Computer Vision – ECCV 2024, Milan, Italy, 29 September–4 October 2024; pp. 430–448. https://doi.org/10.1007/978-3-031-73202-7_25.
21. Yu, F.; Gu, J.; Li, Z.; Hu, J.; Kong, X.; Wang, X.; He, J.; Qiao, Y.; Dong, C. Scaling Up to Excellence: Practicing Model Scaling for Photo-Realistic Image Restoration In the Wild. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 16–22 June 2024; pp. 25669–25680. https://doi.org/10.1109/CVPR52733.2024.02425.
22. Wu, R.; Yang, T.; Sun, L.; Zhang, Z.; Li, S.; Zhang, L. SeeSR: Towards Semantics-Aware Real-World Image Super-Resolution. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 16–22 June 2024; pp. 25456–25467. https://doi.org/10.1109/CVPR52733.2024.02405.
23. Gu, A.; Goel, K.; Ré, C. Efficiently Modeling Long Sequences with Structured State Spaces. In Proceedings of the International Conference on Learning Representations (ICLR), Virtual, 25–29 April 2022.
24. Liu, Y.; Tian, Y.; Zhao, Y.; Yu, H.; Xie, L.; Wang, Y.; Ye, Q.; Jiao, J.; Liu, Y. VMamba: Visual State Space Model. In Proceedings of the

Advances in Neural Information Processing Systems, Vancouver, BC, Canada, 10–15 December 2024; Vol. 37, pp. 103031–103063. https://doi.org/10.52202/079017-3273.
25. Tran, D.P.; Hung, D.D.; Kim, D. VSRM: A Robust Mamba-Based Framework for Video Super-Resolution. In Proceedings of the Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), Honolulu, HI, USA, 18–26 October 2025; pp. 14711–14721.
26. Zamir, S.W.; Arora, A.; Khan, S.; Hayat, M.; Khan, F.S.; Yang, M.H.; Shao, L. Learning Enriched Features for Fast Image Restoration and Enhancement. *IEEE Trans. Pattern Anal. Mach. Intell.* **2023**, *45*, 1934–1948. https://doi.org/10.1109/TPAMI.2022.3167175.
27. Chen, X.; Fan, H.; Girshick, R.; He, K. Improved Baselines with Momentum Contrastive Learning, 2020, [arXiv:cs.CV/2003.04297].
28. Zheng, K.; Zhang, H.; Huang, W. DiffKendall: A Novel Approach for Few-Shot Learning with Differentiable Kendall's Rank Correlation. In Proceedings of the Advances in Neural Information Processing Systems, New Orleans, LA, USA, 10–16 December 2023; Vol. 36, pp. 49403–49415. https://doi.org/10.52202/075280-2149.
29. Yu, L.; Zhang, D.; Liu, N.; Zhou, W. A Multi-View Fusion Method via Tensor Learning and Gradient Descent for Image Features. *IEEE Access* **2021**, *9*, 79389–79399. https://doi.org/10.1109/ACCESS.2021.3079499.
30. Qian, D.; Yu, L.; Tang, H.; Zhao, J. Multiview Feature Fusion Optimization Method for Image Retrieval Based on Matrix Correlation. *J. Electron. Imaging* **2020**, *29*, 053007. https://doi.org/10.1117/1.JEI.29.5.053007.
31. Cui, C.; Liu, L.; Qiao, R. A Cutting-Edge Video Anomaly Detection Method Using Image Quality Assessment and Attention Mechanism-Based Deep Learning. *Alex. Eng. J.* **2024**, *108*, 476–485. https://doi.org/10.1016/j.aej.2024.07.103.
32. Yin, X.; Chen, L. Image Object Detection Method Based on Improved Faster R-CNN. *J. Circuits Syst. Comput.* **2024**, *33*, 2450130. https://doi.org/10.1142/S0218126624501305.
33. Tian, C.; Chen, Y. Image Segmentation and Denoising Algorithm Based on Partial Differential Equations. *IEEE Sens. J.* **2020**, *20*, 11935–11942. https://doi.org/10.1109/JSEN.2019.2959704.
34. Perez, E.; Strub, F.; de Vries, H.; Dumoulin, V.; Courville, A. FiLM: Visual Reasoning with a General Conditioning Layer. *Proc. AAAI Conf. Artif. Intell.* **2018**, *32*, 3942–3951. https://doi.org/10.1609/aaai.v32i1.11671.
35. Kendall, M.G. A New Measure of Rank Correlation. *Biometrika* **1938**, *30*, 81–93. https://doi.org/10.1093/biomet/30.1-2.81.
36. Zhao, H.; Gallo, O.; Frosio, I.; Kautz, J. Loss Functions for Image Restoration With Neural Networks. *IEEE Trans. Comput. Imaging* **2017**, *3*, 47–57. https://doi.org/10.1109/TCI.2016.2644865.
37. Johnson, J.; Alahi, A.; Li, F.F. Perceptual Losses for Real-Time Style Transfer and Super-Resolution. In Proceedings of the Computer Vision – ECCV 2016, Amsterdam, The Netherlands, 8–16 October 2016; pp. 694–711. https://doi.org/10.1007/978-3-319-46475-6_43.
38. Goodfellow, I.; Pouget-Abadie, J.; Mirza, M.; Xu, B.; Warde-Farley, D.; Ozair, S.; Courville, A.; Bengio, Y. Generative Adversarial Nets. In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), Montreal, QC, Canada, 8–13 December 2014; Vol. 27, pp. 2672–2680.
39. Nah, S.; Baik, S.; Hong, S.; Moon, G.; Son, S.; Timofte, R.; Lee, K.M. NTIRE 2019 Challenge on Video Deblurring and Super-Resolution: Dataset and Study. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Long Beach, CA, USA, 16–17 June 2019; pp. 1996–2005. https://doi.org/10.1109/CVPRW.2019.00251.
40. Wang, Z.; Bovik, A.C.; Sheikh, H.R.; Simoncelli, E.P. Image Quality Assessment: From Error Visibility to Structural Similarity. *IEEE Trans. Image Process.* **2004**, *13*, 600–612. https://doi.org/10.1109/TIP.2003.819861.
41. Zhang, R.; Isola, P.; Efros, A.A.; Shechtman, E.; Wang, O. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Salt Lake City, UT, USA, 18–23 June 2018; pp. 586–595. https://doi.org/10.1109/CVPR.2018.00068.
42. Ding, K.; Ma, K.; Wang, S.; Simoncelli, E.P. Image Quality Assessment: Unifying Structure and Texture Similarity. *IEEE Trans. Pattern Anal. Mach. Intell.* **2022**, *44*, 2567–2581. https://doi.org/10.1109/TPAMI.2020.3045810.
43. Mittal, A.; Soundararajan, R.; Bovik, A.C. Making a "Completely Blind" Image Quality Analyzer. *IEEE Signal Process. Lett.* **2013**, *20*, 209–212. https://doi.org/10.1109/LSP.2012.2227726.
44. Mittal, A.; Moorthy, A.K.; Bovik, A.C. No-Reference Image Quality Assessment in the Spatial Domain. *IEEE Trans. Image Process.* **2012**, *21*, 4695–4708. https://doi.org/10.1109/TIP.2012.2214050.
45. Wang, J.; Chan, K.C.; Loy, C.C. Exploring CLIP for Assessing the Look and Feel of Images. *Proc. AAAI Conf. Artif. Intell.* **2023**, *37*, 2555–2563. https://doi.org/10.1609/aaai.v37i2.25353.
46. Radford, A.; Kim, J.W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; et al. Learning Transferable Visual Models from Natural Language Supervision. In Proceedings of the Proceedings of the 38th International Conference on Machine Learning (ICML), Virtual, 18–24 July 2021; Vol. 139, *Proceedings of Machine Learning Research*, pp. 8748–8763.
47. Kingma, D.P.; Ba, J. Adam: A Method for Stochastic Optimization. In Proceedings of the International Conference on Learning Representations (ICLR), San Diego, CA, USA, 7–9 May 2015; [arXiv:cs.LG/1412.6980].
48. Li, D.; Shi, X.; Zhang, Y.; Cheung, K.C.; See, S.; Wang, X.; Qin, H.; Li, H. A Simple Baseline for Video Restoration with Grouped

Spatial-Temporal Shift. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Vancouver, BC, Canada, 17–24 June 2023; pp. 9822–9832. https://doi.org/10.1109/CVPR52729.2023.00947.
49. Antic, J. DeOldify, 2019. GitHub repository; accessed on 1 April 2026.
50. Wan, Z.; Zhang, B.; Chen, D.; Zhang, P.; Chen, D.; Liao, J.; Wen, F. Old Photo Restoration via Deep Latent Space Translation. *IEEE Trans. Pattern Anal. Mach. Intell.* **2023**, *45*, 2071–2087. https://doi.org/10.1109/TPAMI.2022.3163183.
51. Blau, Y.; Michaeli, T. The Perception-Distortion Tradeoff. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Salt Lake City, UT, USA, 18–23 June 2018; pp. 6228–6237. https://doi.org/10.1109/CVPR.2018.00652.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.